INSTITUTE OF COMPUTING & DIGITAL INNOVATION

Capstone Project 2 Plenary Sessions

How to write Chapter 2 - Review of Related Literature and Studies

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Presenter

OUTLINE OF THE PRESENTATION

- Preliminaries – 3 minutes
- **Related Literature and Studies** – 7 minutes
- **System Technical Background** – 7 minutes
- **Synthesis** – 7 minutes
- Sample Chapter 2 – 3 minutes
- Rank of References – 1 minute
- Demonstration of a citation machine using **mybib** – 3 minutes
- Q & A

LEARNING OUTCOMES

- Identify and critically analyze relevant scholarly studies and literature (last 5 years) that support the project's concepts and objectives.

- Document the system's technical background, including all necessary hardware, software, and tools.

- Synthesize findings from various sources, highlighting key interrelations, conflicts, and, most importantly, identifying the **research gap** that the proposed IS project will fill.

SDG & KLD REGAL 100 INTEGRATION

- **SDG 17: Partnerships for the Goals.**

  The module encourages the review of global academic literature, fostering a broader, collaborative knowledge base.

- **KLD REGAL100: Responsibility & Ethics.**

  Students demonstrate **ethical responsibility** by accurately citing and acknowledging original authors, ensuring academic integrity, and avoiding duplication of existing work.

WHY IS CHAPTER 2 IMPORTANT?

- It establishes the **academic and technical foundation** for your Information System solution.

- The purpose of the literature review is twofold:
  - to demonstrate that you understand the existing body of knowledge;
  - to justify the necessity and originality of your project by identifying a **gap** that needs to be addressed.

1. Related Literature and Studies

- This section presents a thorough review of published and unpublished works (journal articles, conference papers, theses) related to your project's concept.
- **Focus on Relevance:** Only include literature that is directly related to your project's technology, methodology, or problem domain.
- **Comparison and Contrast:** Do not just summarize. You must actively **compare** the findings, methodologies, scope, and limitations of the previous studies. Highlight similarities (where they support your concepts) and differences (where they show a variation in approach).
- **Time Frame:** Focus on research conducted over the last **five years** to ensure the information is current and relevant to modern IS practices.

2. Technical System Background

- This is where you detail the technologies that will be used to build your proposed system. This section validates your technical choices by linking them to established IT practice.
- **Key Components:** Provide an overview and detailed explanation of:
- **Programming Languages and Frameworks:** (e.g., Python, React, Laravel) and why they are best suited for the system's requirements (e.g., speed, scalability).
- **Databases:** (e.g., MySQL, MongoDB) and the specific reasons for selection (e.g., relational vs. non-relational).
- **Hardware and Tools:** (e.g., server requirements, specific sensors if IoT is involved, APIs).
- **Architecture:** Briefly outline the proposed system's architectural style (e.g., three-tier architecture, microservices) and the technical standards that will guide development.

3. Synthesis

- The synthesis is the integrating conclusion of your literature review. It connects all the dots.
- **Integration:** Combine key insights, theories, and concepts from the review, showing how they collectively inform your project's design.
- **The Research Gaps (Crucial):** This is the most important element. Based on your review, clearly state the **gaps, conflicts, or unanswered questions** in the existing knowledge. You must then explicitly explain how your Capstone Project—specifically, the features or methodologies you propose will fill that gap.

  Example: "While existing studies address inventory management (Studies A and B), no one integrates a **real-time AI-driven demand forecasting module** specifically optimized for perishable goods in a decentralized retail environment, which is the precise gap this system will fill."

Sample Chapter 2: Review of Related Literature and Studies

1. Related Literature and Studies

This section reviews scholarly articles and capstone studies related to alumni systems, data management, and business intelligence to establish the foundation and context for the proposed system.

1.1 The Evolution of Alumni Information Systems

Smith and Davies (2023) developed a **mobile-first Alumni Engagement Platform** aimed at improving communication and event participation. Their methodology emphasized high usability and social integration, successfully increasing event attendance by 40%. However, their system focused primarily on engagement and **lacked a robust data governance or BI module** for statistical reporting. Conversely, Chen et al. (2020) implemented a **desktop-based Alumni Data Warehouse** to centralize all historical academic records. While successful in data standardization, their approach relied on manual data migration and batch updates, resulting in data latency.

Continuation…

- **Comparison:** Smith and Davies's system prioritized real-time user interaction (supporting the *engagement* aspect of the proposed system), while Chen et al.'s work prioritized data standardization (supporting the *records management aspect*). Both studies, however, reveal a common limitation: a trade-off between real-time data input (mobile engagement) and strategic data analysis (BI).

Continuation…

- **1.2 The Application of Business Intelligence (BI) in Education**

  According to the findings of Garcia (2021), the utilization of **Business Intelligence dashboards** in university administration has been instrumental in transforming raw student data into **actionable insights** for curriculum planning and resource allocation. Garcia's methodology showed that interactive BI visualizations reduced the time needed for executive decision-making by 60%. Similarly, Lee and Kim (2022) explored BI integration specifically within **university reporting systems**. They concluded that predefined static reports fail to meet dynamic compliance requirements, advocating for systems that allow users to customize their data views.

  Contrast: The research consistently supports the high value of BI in education. However, Garcia's work focused on student enrollment and performance, while Lee and Kim focused on curriculum. Neither study explicitly detailed the implementation of BI metrics tailored for **alumni employment rate tracking** or the specific statistical requirements of government accreditation bodies—a critical need of the current study's client.

Continuation…

- **1.3 Technical Frameworks for Web-Based Development**

  Dela Cruz (2020) compared various technology stacks for developing educational IS, concluding that the **MERN (MongoDB, Express, React, Node.js) stack** provided the best balance between performance, scalability, and ease of maintenance for systems handling moderate user loads. This choice ensures a fast, asynchronous user experience. In contrast, Johnson (2019) utilized a traditional LAMP stack for an internal records system. While reliable, the report noted significant performance bottlenecks and slower development cycles when integrating advanced visualization libraries compared to modern JavaScript frameworks.

  Relevance: This literature supports the modern approach to system development. The proposed system will leverage contemporary JavaScript frameworks (like React/MERN, following Dela Cruz's findings) to ensure the system is **scalable and responsive**, contrasting with the slower performance experienced by Johnson's study.

Model Chapter 2: Review of Related Literature and Studies

2. Technical System Background

The proposed **Web-Based Alumni Tracer and Records Management System** require a secure, scalable, and modern technical foundation to ensure longevity and data integrity.

2.1 Programming Environment and Frameworks: The system will be developed using **JavaScript** within the **MERN (MongoDB, Express, React, Node.js) stack**.

Node.js (Backend): Selected for its non-blocking I/O model, which is essential for handling multiple concurrent alumni registrations and data updates without lag.

React (Frontend): Chosen for its component-based architecture, which facilitates modular development, improves code reusability, and enables a highly responsive user interface necessary for both administrators and alumni.

2.2 Database Management System (DBMS): MongoDB will be used as the primary NoSQL database. This selection is based on its flexibility, which is ideal for storing varied alumni employment profiles, academic records, and unstructured tracer survey responses efficiently.

Continuation…

2.3 Business Intelligence (BI) Tools: The BI integration will rely on a dedicated JavaScript data visualization library, such as **D3.js or Recharts**. These tools will dynamically connect to the MongoDB data layer to generate interactive charts and graphical reports, fulfilling the need for strategic visualization identified in the literature review (Garcia, 2021).

2.4 Deployment Environment: The system will be deployed on a **cloud-based server** (e.g., AWS or Google Cloud) to ensure high availability, security, and scalability to accommodate the University's large volume of alumni data.

Model Chapter 2: Review of Related Literature and Studies

3. Synthesis

The reviewed literature confirms the need for an efficient digital solution for alumni management and strongly validates the strategic advantage of integrating BI tools into administrative functions (Garcia, 2021; Lee & Kim, 2022). Previous studies successfully addressed either **alumni engagement** (Smith & Davies, 2023) or **historical data standardization** (Chen et al., 2020). However, a significant gap remains: there is a noticeable lack of published literature detailing the integration of a **real-time, MERN-stack based system** that can simultaneously handle high volumes of user-generated profile updates *and* perform dynamic BI analytics specifically focused on **accreditation-critical metrics like alumni employment tracing**.

Continuation…

- **The Research Gaps:** Existing systems often sacrifice real-time analytics for comprehensive records, or vice-versa. The gap, therefore, is the absence of a comprehensive, modern web-based solution that **seamlessly connects dynamic, user-input tracer data to a BI module** for immediate, strategic decision-making and reporting, optimized for the Philippine educational context.
- **How the Project Fills the Gaps:** By developing the **Web-Based Alumni Tracer and Records Management System with BI Integration**, this study aims to bridge this gap. It integrates the robust records management practices suggested by Chen et al. with a responsive, modern framework (Dela Cruz, 2020) to deliver the critical, real-time BI reports necessary to satisfy the specific accreditation and strategic planning needs of the client institution, a feature not comprehensively covered in the reviewed literature.

Plenary Session 2

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Additional Information

1. High-Quality/Primary Sources (Most Preferred)

These sources provide direct, first-hand evidence or original research. LibGuides

- **Peer-Reviewed Journal Articles:** Research published in academic journals that has been evaluated by experts in the field.
- **Theses and Dissertations:** Original research conducted for university degrees, usually unpublished but highly detailed.
- **Government Documents & Reports:** Official data,, laws, and, reports from agencies like WHO, UNESCO, or the DepEd.
- **Conference Proceedings:** Papers presented at academic conferences, often containing the latest, cutting-edge research.
- **Original Data/Artifacts:** Raw data, diaries, letters, photographs, or witness reports. University of California San Diego +6

2. Secondary Sources (Highly Acceptable)

These discuss, interpret, or analyze information from primary sources. LibGuides

- **Review Articles & Scholarly Books:** Books and articles that analyze or synthesize existing research.
- **Professional Magazines/Journals:** Reputable, expert-written articles (e.g., *Harvard Business Review*).
- **Textbooks:** Academic books that summarize research in a specific field. BCcampus Pressbooks +4

3. Tertiary Sources (For Background Information)

These collect, summarize, and reorganize primary and secondary sources. They are excellent for finding background information but are rarely used as direct evidence in high-level research. University of Michigan-Flint +4

- **Encyclopedias and Dictionaries:** e.g., Wikipedia, Subject Encyclopedias.
- **Almanacs and Bibliographies:**.
- **Indexes and Search Engines:** Databases like Google Scholar.

4. Grey Literature & Questionable Sources (Use with Caution)

- **Working Papers/Preprints:** Draft versions of papers (useful for latest research but not yet peer-reviewed).
- **Popular News/Magazines:** Newspaper reports, articles for the general public.
- **Unverified Internet Sources:** Blogs or non-academic websites. YouTube +4